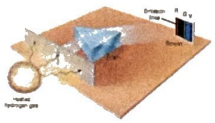
## The Big Question

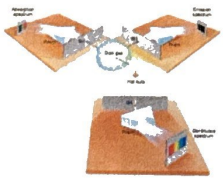
Atoms are the basic components of normal matter—the stuff from which stars, planets, and we ourselves are built. In turn, atoms are made of smaller elementary particles: protons, neutrons, and electrons. But even these particles are not fundamental, for physicists know that protons and neutrons are made of quarks, and many theorists think that quarks and electrons may have even deeper structure. How far does this hierarchy continue, and what new windows on the cosmos will be opened by our understanding of these unknown levels of nature?

# Chapter Review

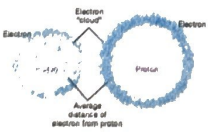
## SUMMARY



1 A **spectroscope (p. 80)** is a device for splitting a beam of radiation into its component frequencies for detailed study. Many hot objects emit a **continuous spectrum (p. 80)** of radiation, containing light of all wavelengths. A hot gas may instead produce an **emission spectrum (p. 81),** consisting of only a few well-defined **emission lines (p. 81)** of specific frequencies, or colors. Passing a continuous beam of radiation through cool gas will produce **absorption lines (p. 81)** at precisely the same frequencies as would be present in the gas's emission spectrum.

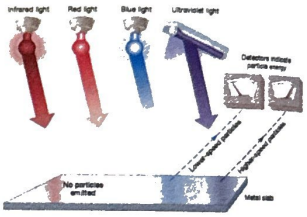


2 **Kirchhoff's laws (p. 83)** describe the relationships among these different types of spectra. The emission and absorption lines produced by each element are unique—they provide a "fingerprint" of that element. The study of the spectral lines produced by different substances is called **spectroscopy (p. 83).** Spectroscopic studies of the Fraunhofer lines in the solar spectrum yield detailed information about the Sun's composition.

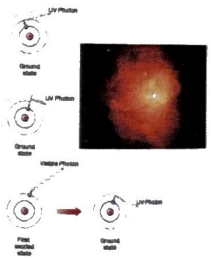



3 **Atoms (p. 85)** are made up of negatively charged electrons orbiting a positively charged **nucleus (p. 85)** consisting of positively charged protons and electrically neutral **neutrons (p. 90).** The number of protons in the nucleus determines the particular **element (p. 90)** the atom represents. In the **Bohr model (p. 86),** a hydrogen atom has a minimum energy **ground state (p. 86)** representing its "normal" condition. When the electron has a higher than normal energy, the atom is in an **excited state (p. 86).** For any given atom, only certain, well-defined energies are possible. In the modern view, the electron is envisaged as being spread out in a "cloud" around the nucleus, but still with a sharply defined energy.

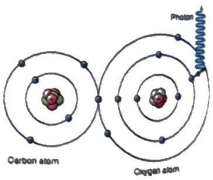
4 Electromagnetic radiation exhibits both wave and particle properties. Particles of radiation are called **photons (p. 86).**



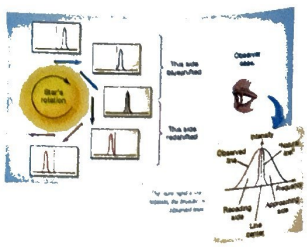

In order to explain the **photoelectric effect (p. 85),** Einstein found that the energy of a photon must be directly proportional to the photon's frequency.




5 As electrons move between energy levels within an atom, the difference in energy between the states is emitted or absorbed in the form of photons. Because the energy levels have definite energies, the photons also have definite energies, and hence colors, that are characteristic of the type of atom involved.




6 **Molecules (p. 91)** are groups of two or more atoms bound together by electromagnetic forces. Like atoms, molecules exist in energy states that obey rules similar to those governing the internal structure of atoms. When a molecule makes a transition between energy states, it emits or absorbs a characteristic spectrum of radiation that identifies it uniquely.




7 Astronomers apply the laws of spectroscopy to analyze radiation from beyond Earth. Several physical mechanisms can broaden spectral lines. The most important is the Doppler effect, which occurs because stars are hot, or rotating, or turbulent, so their atoms are in constant motion.

## Discussion

MasteringAstronomy® For instructor-assigned homework go to MasteringAstronomy.

Problems labeled **POS** explore the process of science **VIS** problems focus on reading and interpreting visual information. **LO** connects to the introduction's numbered Learning Outcomes. **BP** builds on concepts in the Big Picture.

1. **LO1** What is an absorption spectrum? An emission spectrum? How are they related?
2. **LO1 LO2 BP** What is spectroscopy? How can spectroscopy be used to infer the composition and temperature of a star?
3. **LO2 BP** Describe the basic components of a simple spectroscope and the additional components in spectrographs used by astronomers for modern observations.
4. **LO4** Why are gamma rays generally harmful to life-forms, but radio waves are generally harmless?
5. **LO4** In the particle description of light, what is color?
6. **LO3** What is an atom? In what ways does the Bohr model of atomic structure differ from the modern view?
7. **LO3** Give a brief description of a hydrogen atom.
8. **LO4** What does it mean to say that a physical quantity is quantized? Why do we think light is quantized?
9. **LO5** What is the normal condition for atoms? What is an excited atom? What are orbitals?
10. **LO5** Why do excited atoms absorb and reemit radiation at characteristic frequencies?
11. **LO5 BP POS** How are absorption and emission lines produced in a stellar spectrum? What information might absorption lines in the spectrum of a star reveal about a cloud of cool gas lying between us and the star?
12. **LO7 BP** Why might spectral lines of an element in a star's spectrum be weak, even though that element is abundant in the star?
13. **LO6** How do molecules produce spectral lines unrelated to the movement of electrons between energy levels?
14. **LO7** How can the Doppler effect broaden a spectral line?
15. **LO7 BP POS** List three properties of a star that can be determined from observations of its spectrum.

## Conceptual Self-Test: Multiple Choice

1. **LO1** Compared with a ground-based spectrum, the spectrum of a star observed from above Earth's atmosphere would show **(a)** no absorption lines; **(b)** fewer emission lines; **(c)** fewer absorption lines; **(d)** many more absorption lines.
2. **LO1** The visible spectrum of sunlight reflected from Saturn's cold moon Titan would be expected to be **(a)** continuous; **(b)** an emission spectrum; **(c)** an absorption spectrum.
3. **LO1 VIS** Figure 4.3 ("Elemental Emission") shows the emission spectrum of neon gas. If the temperature of the gas were increased, we would observe **(a)** fewer red lines and more blue lines; **(b)** even more red lines; **(c)** some faint absorption features; **(d)** no significant change.
4. **LO2** Compared with a star having many blue absorption lines, a star with many red and blue absorption lines must be **(a)** cooler; **(b)** of different composition; **(c)** moving away from the observer; **(d)** moving away from the other star.
5. **LO3** An atom that has been ionized **(a)** has equal numbers of protons and electrons; **(b)** has more protons than electrons; **(c)** is radioactive; **(d)** is electrically neutral.
6. **LO5 VIS** In Figure 4.10 ("Atomic Excitation"), compared with an electron transition from the first excited state to the ground state, a transition from the third excited state to the second excited state emits a photon of **(a)** greater energy; **(b)** lower energy; **(c)** identical energy.
7. **LO5** Compared with a complex atom like neon, a simple atom such as hydrogen has **(a)** more excited states; **(b)** fewer excited states; **(c)** the same number of excited states.
8. **LO7** Compared with cooler stars, the hottest stars have absorption lines that are **(a)** thin and distinct; **(b)** broad and fuzzy; **(c)** identical to the lines in the cooler stars.
9. **LO7** Compared with slowly rotating stars, the fastest spinning stars have absorption lines that are **(a)** thin and distinct; **(b)** broad and fuzzy; **(c)** identical to the lines in the slowly rotating stars.
10. **LO7** Astronomers analyze starlight to determine a star's **(a)** temperature; **(b)** composition; **(c)** motion; **(d)** all of the above.

# Problems

*The number of dots preceding each Problem indicates its approximate level of difficulty.*

1. • What is the energy (in electron volts—see *More Precisely 4-1)* of a 450-nm blue photon? A 200-nm ultraviolet photon?
2. • What is the energy (in electron volts) of a 100-GHz (1 gigahertz = $10^9$ Hz) microwave photon?
3. • What are the wavelengths of (a) a 2-eV red photon, (b) a 0.1-eV infrared photon, and (c) a 5000-eV (5-keV) X-ray?
4. • How many times more energy does a 1-nm gamma ray have than a 10-MHz radio photon?
5. •• List all the spectral lines of hydrogen that lie in the visible range (taken to run from 400 to 700 nm in wavelength).
6. •• How many different photons (i.e., photons of different frequencies) can be emitted as a hydrogen atom in the third excited state falls back, directly or indirectly, to the ground state? What are their wavelengths?
7. • A distant galaxy is receding from Earth with a radial velocity of 3000 km/s. At what wavelength would its Ly$\alpha$ line be received by a detector above Earth's atmosphere?
8. •• In a demonstration of the photoelectric effect, suppose that a minimum energy of $5 \times 10^{-19}$ J (3.1 eV) is required to dislodge an electron from a metal surface. What is the minimum frequency (and longest wavelength) of radiation for which the detector registers a response?

# Activities

## Collaborative

1. Find a spectrum of the Sun that also has a wavelength scale on it. Google is a good place to start. Select some absorption lines and determine their wavelengths by interpolation. Now, try to identify the element that produced these lines. Use a reference of lines such as Moore's "A Multiplet Table of Astrophysical Interest," available on the NASA Astrophysics Data System. Work with the darkest lines before trying the fainter ones. How many elements can you find?
2. Develop a physical model to illustrate how electron transitions within an atom produce a spectrum of absorption or emission lines. How can you represent atomic energy levels, electrons, photons, and their interactions? Use your model to demonstrate Kirchhoff's Laws.

## Individual

1. Obtain a handheld spectroscope, available from your school science lab or online. In the shade, point the spectroscope at a white cloud or white piece of paper that is in direct sunlight. Look for the absorption lines in the Sun's spectrum. Note their wavelength from the scale inside the spectroscope. How many of the lines can you identify by comparing your list with the Fraunhofer lines given in many astronomy reference books, or on Wikipedia?
2. Obtain a handheld spectroscope or diffraction grating, a lamp, and several types of light bulbs (incandescent, florescent, neon, LED, etc.). Observe the spectrum produced by each light bulb and describe what you see. What type of spectrum is produced by each light bulb? How is each spectrum generated?